

计算机设计与实践

流水线CPU及SoC设计

联合调试

2026·夏



HITSZ 实验与创新实践教育中心
Education Center of Experiments and Innovations, HITSZ

联合调试

- ◆ 组员甲完成流水线CPU的Basic Trace调试
- ◆ 组员乙完成单周期SoC的AXI Trace及C_TEST的下板测试
- ◆ 把流水线CPU集成到SoC，完成AXI Trace测试
- ◆ 完成流水线SoC的C_TEST下板测试
- ◆ 在流水线SoC上运行CoreMark或LLAMA2推理程序
- ◆ 优化性能、下板验收
- ◆ **注意代码规范性，可以避免很多问题！**



优化方法简介

◆ 主要优化角度

1. CPU频率

- **流水线CPU主频**取决于延迟最大的流水级
 - 关键路径：寄存器与寄存器之间延迟最大的信号传递路径
 - 关键路径延迟越小越好

2. 访存性能

- 访存性能取决于：物理介质、总线效率、Cache性能

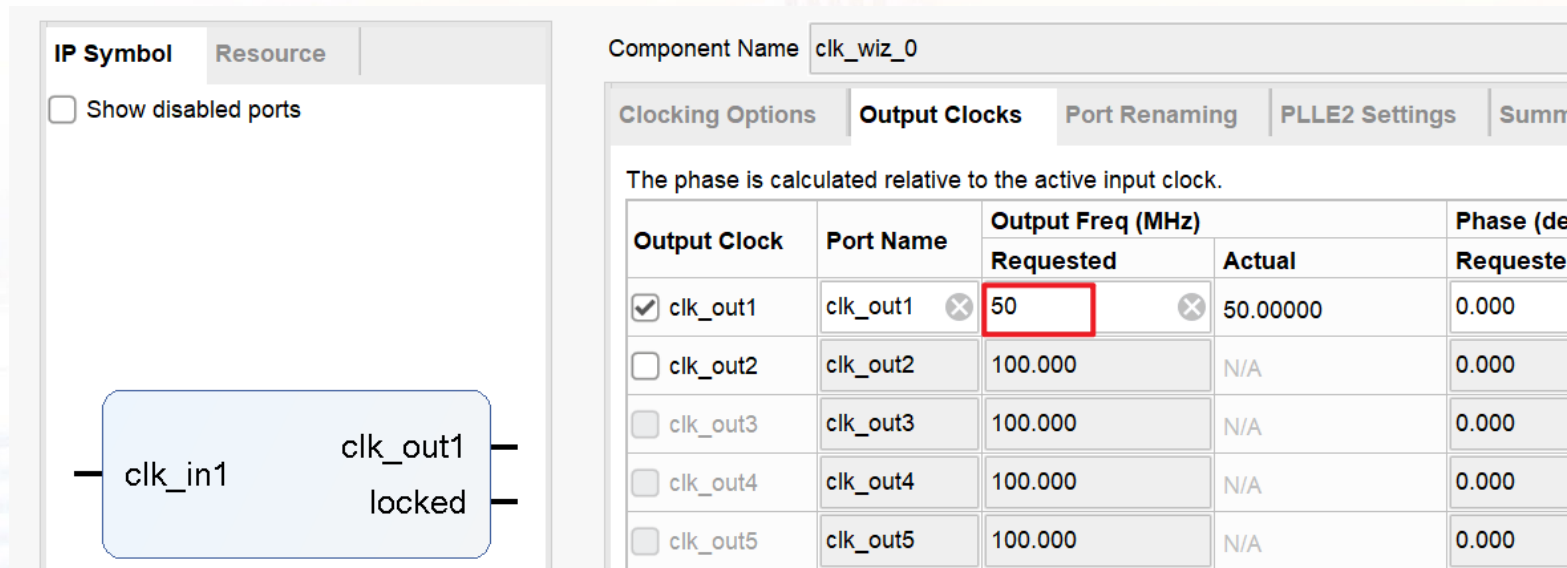
3. 运算性能

- 提升运算的并行性，比如向量运算MAC4指令、多发射， etc.

关键路径

◆ 借助Vivado寻找关键路径

① 更改时钟IP核设置，提高CPU主时钟频率至期望值



IP Symbol Resource

☐ Show disabled ports

Component Name clk_wiz_0

Clocking Options Output Clocks Port Renaming PLLE2 Settings Summary

The phase is calculated relative to the active input clock.

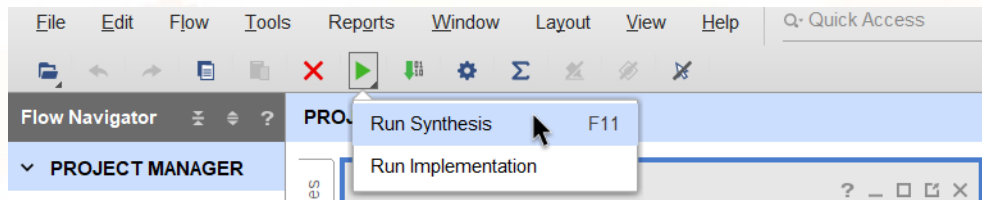
Output Clock	Port Name	Output Freq (MHz)		Phase (deg)
		Requested	Actual	
☒ clk_out1	clk_out1	50	50.00000	0.000
☐ clk_out2	clk_out2	100.000	N/A	0.000
☐ clk_out3	clk_out3	100.000	N/A	0.000
☐ clk_out4	clk_out4	100.000	N/A	0.000
☐ clk_out5	clk_out5	100.000	N/A	0.000

clk_in1 clk_out1 locked

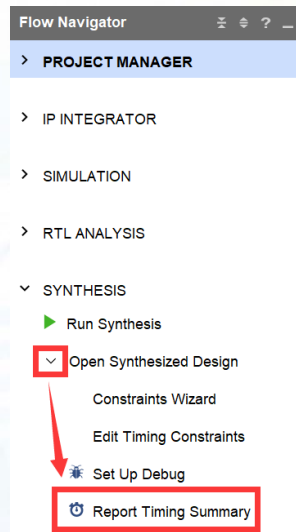
关键路径

◆ 借助Vivado寻找关键路径

② F11或操作GUI运行综合



③ 综合成功后，查看Timing Summary报告

A screenshot of the 'Design Timing Summary' report in Vivado. The report is divided into three columns: Setup, Hold, and Pulse Width. In the Setup column, the 'Total Negative Slack (TNS)' is highlighted with a red circle and a purple arrow pointing to the text '存在时序违例，即关键路径延迟大于时钟周期时间'. Below this, the 'Number of Failing Endpoints' is 32. In the Hold column, the 'Worst Hold Slack (WHS)' is 0.115 ns. In the Pulse Width column, the 'Worst Pulse Width Slack (WPWS)' is 2.750 ns. The report also shows 'Timing constraints are not met.'

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): -1.964 ns	Worst Hold Slack (WHS): 0.115 ns	Worst Pulse Width Slack (WPWS): 2.750 ns
Total Negative Slack (TNS): -44.242 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 32	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 8338	Total Number of Endpoints: 8338	Total Number of Endpoints: 8993

关注标红路径

存在时序违例，即关键路径延迟大于时钟周期时间

关键路径

◆ 借助Vivado寻找关键路径

④ 查看关键路径

Tcl Console					Messages	Log	Reports	Design Runs	Timing													
Q										Intra-Clock Paths - clk_out1_cpucik - Setup												
General Information										Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock
Timer Settings										Path 1	-1.964	16	17	44	Core_cpu/Reg_ID2EX/alu_a_ex_reg[31]/C	Core_cpu/U_PC/pc_o_reg[31]/D	9.740	3.397	6.343	8.0	clk_out1_cpucik	clk_out1_cpucik
Design Timing Summary										Path 2	-1.887	16	17	44	Core_cpu/Reg_ID2EX/alu_a_ex_reg[31]/C	Core_cpu/U_PC/pc_o_reg[30]/D	9.663	3.316	6.347	8.0	clk_out1_cpucik	clk_out1_cpucik
Clock Summary (3)										Path 3	-1.847	15	16	44	Core_cpu/Reg_ID2EX/alu_a_ex_reg[31]/C	Core_cpu/U_PC/pc_o_reg[27]/D	9.623	3.280	6.343	8.0	clk_out1_cpucik	clk_out1_cpucik
Check Timing (69)										Path 4	-1.833	16	17	44	Core_cpu/Reg_ID2EX/alu_a_ex_reg[31]/C	Core_cpu/U_PC/pc_o_reg[29]/D	9.609	3.402	6.207	8.0	clk_out1_cpucik	clk_out1_cpucik
Intra-Clock Paths										Path 5	-1.770	15	16	44	Core_cpu/Reg_ID2EX/alu_a_ex_reg[31]/C	Core_cpu/U_PC/pc_o_reg[26]/D	9.546	3.199	6.347	8.0	clk_out1_cpucik	clk_out1_cpucik
clk_out1_cpucik										Path 6	-1.730	14	15	44	Core_cpu/Reg_ID2EX/alu_a_ex_reg[31]/C	Core_cpu/U_PC/pc_o_reg[23]/D	9.506	3.163	6.343	8.0	clk_out1_cpucik	clk_out1_cpucik
Setup -1.964 ns (10)										Path 7	-1.716	16	17	44	Core_cpu/Reg_ID2EX/alu_a_ex_reg[31]/C	Core_cpu/U_PC/pc_o_reg[28]/D	9.492	3.286	6.206	8.0	clk_out1_cpucik	clk_out1_cpucik
Hold 0.115 ns (10)										Path 8	-1.716	15	16	44	Core_cpu/Reg_ID2EX/alu_a_ex_reg[31]/C	Core_cpu/U_PC/pc_o_reg[25]/D	9.492	3.285	6.207	8.0	clk_out1_cpucik	clk_out1_cpucik
Pulse Width 2.750 ns (31)										Path 9	-1.653	14	15	44	Core_cpu/Reg_ID2EX/alu_a_ex_reg[31]/C	Core_cpu/U_PC/pc_o_reg[22]/D	9.429	3.082	6.347	8.0	clk_out1_cpucik	clk_out1_cpucik
clkfbout_cpucik										Path 10	-1.613	13	14	44	Core_cpu/Reg_ID2EX/alu_a_ex_reg[31]/C	Core_cpu/U_PC/pc_o_reg[19]/D	9.389	3.046	6.343	8.0	clk_out1_cpucik	clk_out1_cpucik
fpga_clk																						

关键路径

◆ 借助Vivado寻找关键路径

④ 查看关键路径：右键点击Schematic可在电路图上查看关键路径

The screenshot displays the Vivado IDE interface with three main components:

- Path Properties Window (Top Left):** Shows details for Path 1, including Name, Slack (-1.964ns), Source (Core_cpuReg_ID2X), Destination (Core_cpuU_PCpc_c), Path Group (clk_out1_cpuck), Path Type (Setup), Requirement (8.000ns), Data P_Delay (9.740ns), Logic Levels (16), Clock ... Skew (-0.145ns), and Clock U_tainty (0.123ns).
- Circuit Schematic (Top Right):** A block diagram showing a sequence of logic blocks connected in a chain, representing the critical path.
- Timing Summary Report (Bottom):** A table listing various timing paths and their characteristics.

Timing Summary - timing_1

Name	Slack	Levels	Routes	High Fanout	Frequency	Mark
Path 1	-1.964	16	17	44	Co	
Path 2	-1.887	16	17	44	Co	
Path 3	-1.847	15	16	44	Co	
Path 4	-1.833	16	17	44	Co	
Path 5	-1.770	15	16	44	Co	
Path 6	-1.730	14	15	44	Co	
Path 7	-1.716	16	17	44	Co	
Path 8	-1.716	15	16	44	Co	
Path 9	-1.653	14	15	44	Co	
Path 10	-1.613	13	14	44	Co	

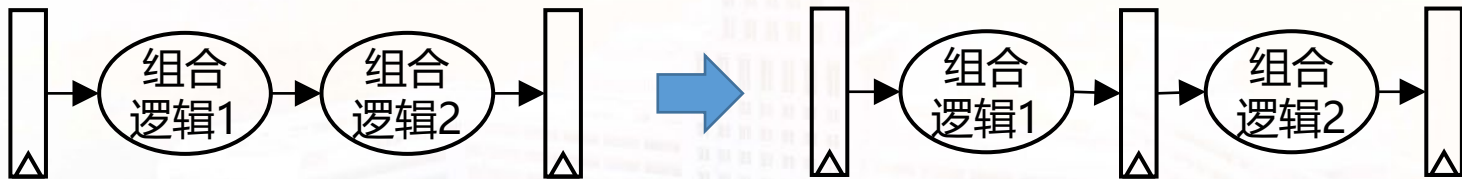
Timing Summary - timing_1 (Table 2)

	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
g[31]D	9.740	3.397	6.343	8.0	clk_out1_cpuck	clk_out1_cpuck		0.123
g[30]D	9.663	3.316	6.347	8.0	clk_out1_cpuck	clk_out1_cpuck		0.123
g[27]D	9.623	3.280	6.343	8.0	clk_out1_cpuck	clk_out1_cpuck		0.123
g[29]D	9.609	3.402	6.207	8.0	clk_out1_cpuck	clk_out1_cpuck		0.123
g[26]D	9.546	3.199	6.347	8.0	clk_out1_cpuck	clk_out1_cpuck		0.123
g[23]D	9.506	3.163	6.343	8.0	clk_out1_cpuck	clk_out1_cpuck		0.123
g[28]D	9.492	3.286	6.206	8.0	clk_out1_cpuck	clk_out1_cpuck		0.123
g[25]D	9.492	3.285	6.207	8.0	clk_out1_cpuck	clk_out1_cpuck		0.123
g[22]D	9.429	3.082	6.347	8.0	clk_out1_cpuck	clk_out1_cpuck		0.123
g[19]D	9.389	3.046	6.343	8.0	clk_out1_cpuck	clk_out1_cpuck		0.123

关键路径

◆ 缩短关键路径

① 在组合逻辑中间插入寄存器，即新增流水级



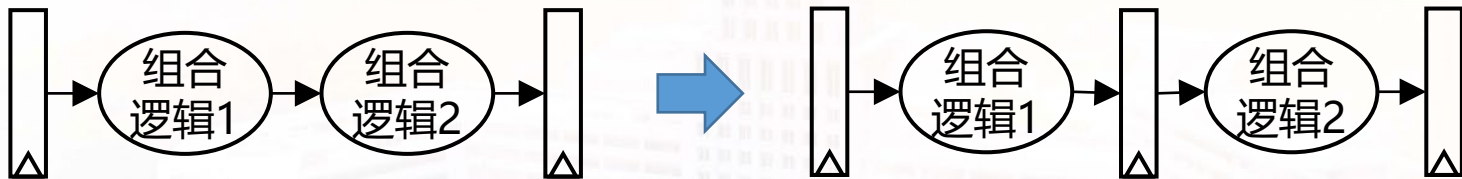
② 平衡不同流水级的延迟



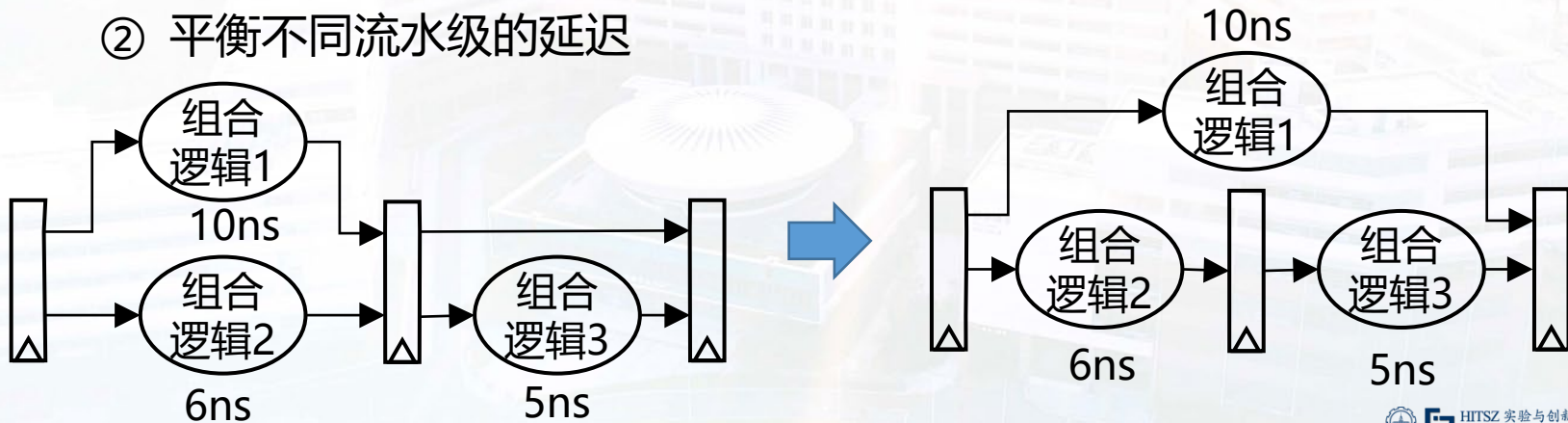
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① 在组合逻辑中间插入寄存器，即新增流水级



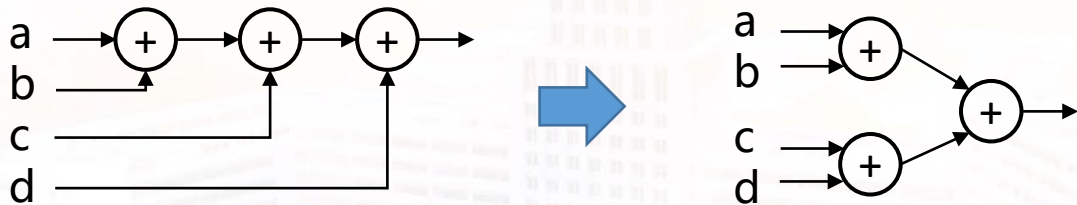
② 平衡不同流水级的延迟



关键路径

◆ 缩短关键路径

③ 缩短操作延迟



➤ 串行改并行，比如多重if-else改用case语句

④ 减小扇出 (fan out)：扇出越大，布线越困难，路劲变长则延迟变大





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